

Air levitation control system

Stage 3

Done by:

Mohamed Khaled 202400729

Khaled Yasser 202402584

Mohamed Sharkawy 202301006

Marwan Mohamed 202201652

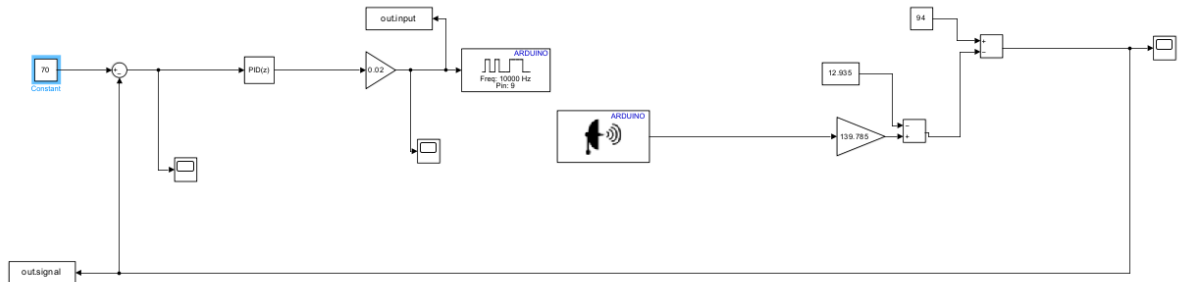
Ahmed Mohamed 202100247

Under the Supervision of:

Dr. Ahmed S. Abd-Rabou

Zewail City of Science and Technology
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Simulink implementation:



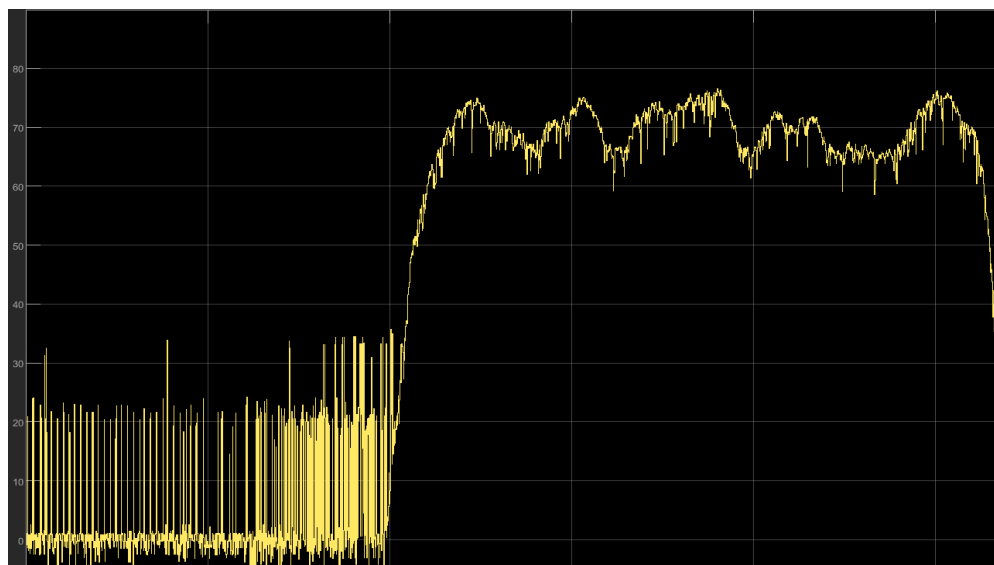
-We implemented our predesigned Controller in the Z domain by taking feedback from the output (distance after calibration).

Testing the Controller in Simulink:

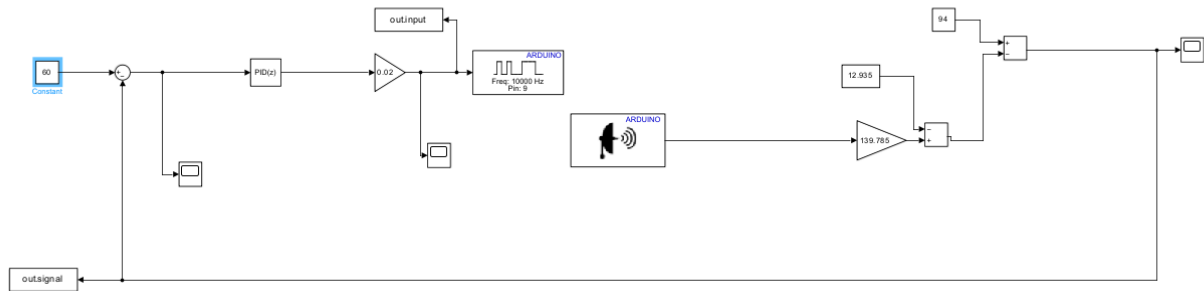
-We will enter the set point (target height) to be some number, so we are expecting the controller to make the system stabilize at this point with almost no change.

-Once any change in position happens suddenly due to any unexpected changes, the Ultrasonic sensor will send feedback to the controller to adjust the PWM input and return to the target height as a result.

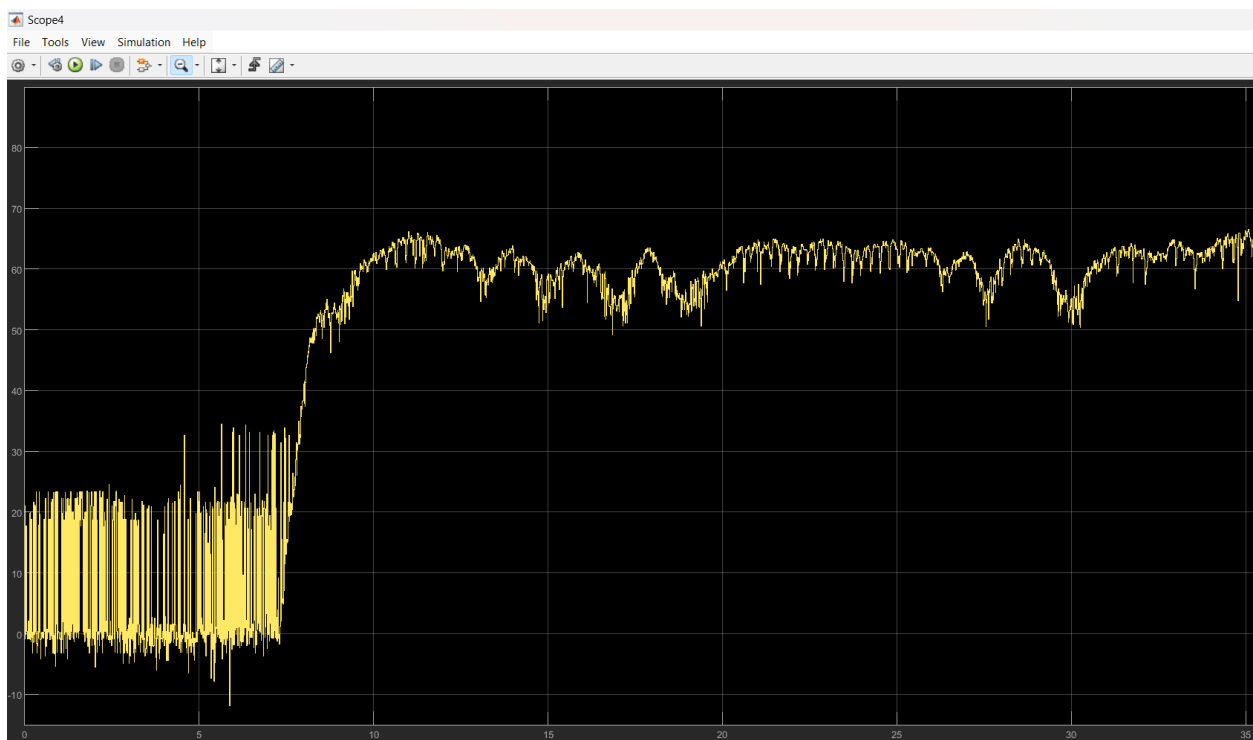
70 cm set point output:



60 cm set point configuration and output:

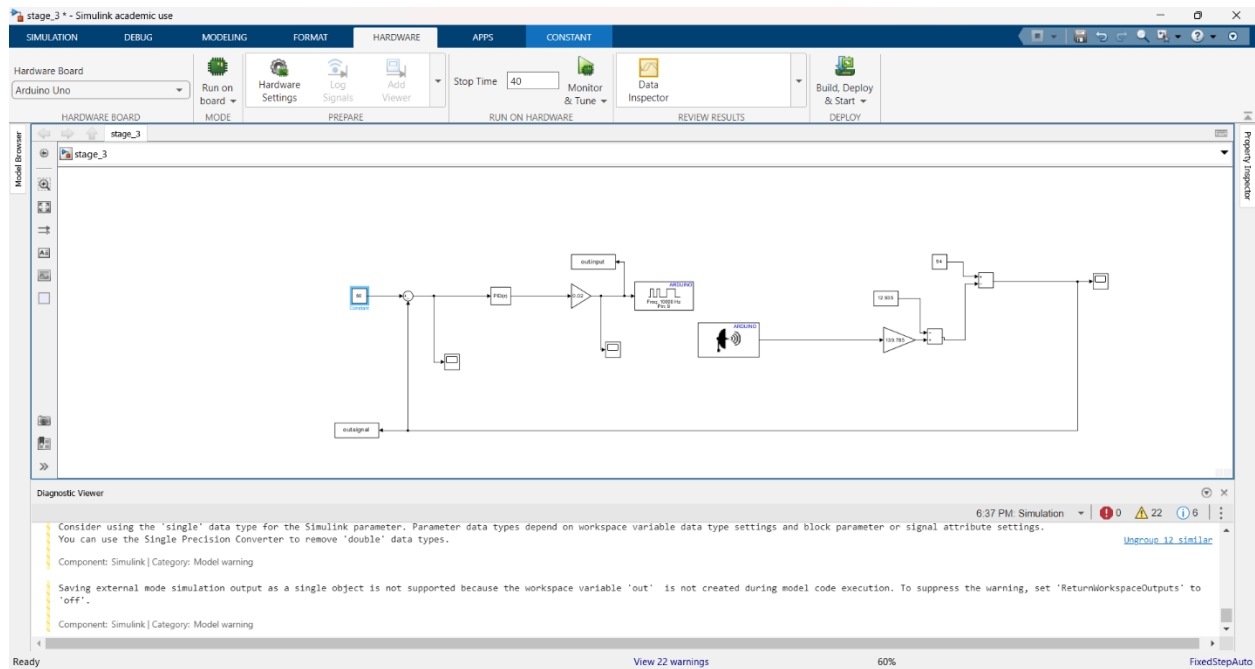


Output:



We can notice that the output is almost at the target position(60 cm),which is a very good result

50 cm set point configuration and output:

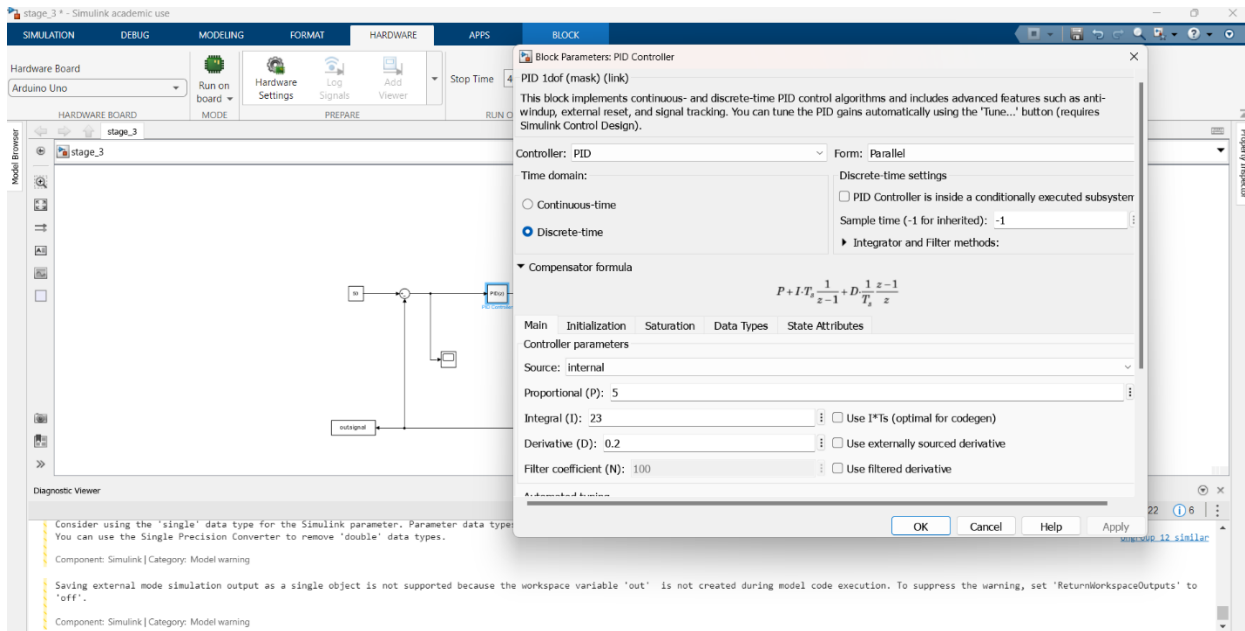


Output:



For the three cases, the output is almost constant at the input set point (reference input), so our controller actually achieves perfect results.

Z-Domain PID Controller parameters and functionality:



$K_p=5$, $K_i=23$ and $K_d=.2$

But we added a gain block=0.02. So it the same as **$K_p=0.1$, $K_i=0.46$ and $K_d=.004$**

With no gain block.

Applying a sudden change to the system:

We will suddenly shut of the power supply to monitor the behavior of the controller.the sensor will read a lower position.So the controller should send a greater PWM value to compensate this sudden decrease.

-Now will test this on Simulink and see the results. And we are expecting the following:

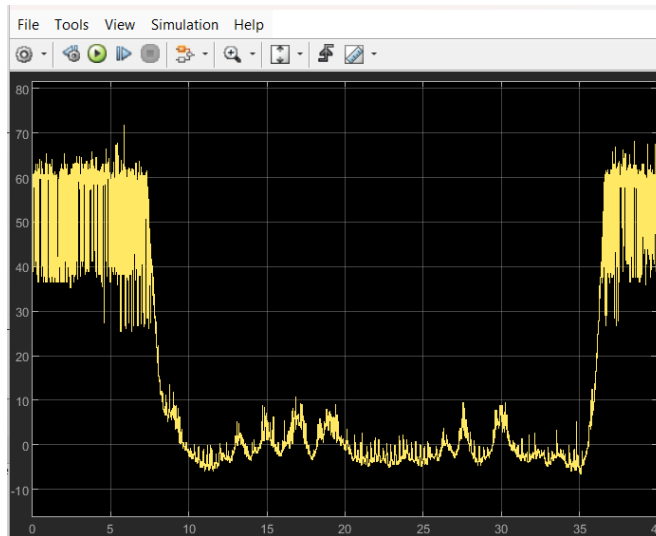
-Ultrasonic sensor reading will decrease.

- The input in the scope just before the controller will suddenly increase (representing an increase in the error).

-The output in the scope after the controller will suddenly increase.

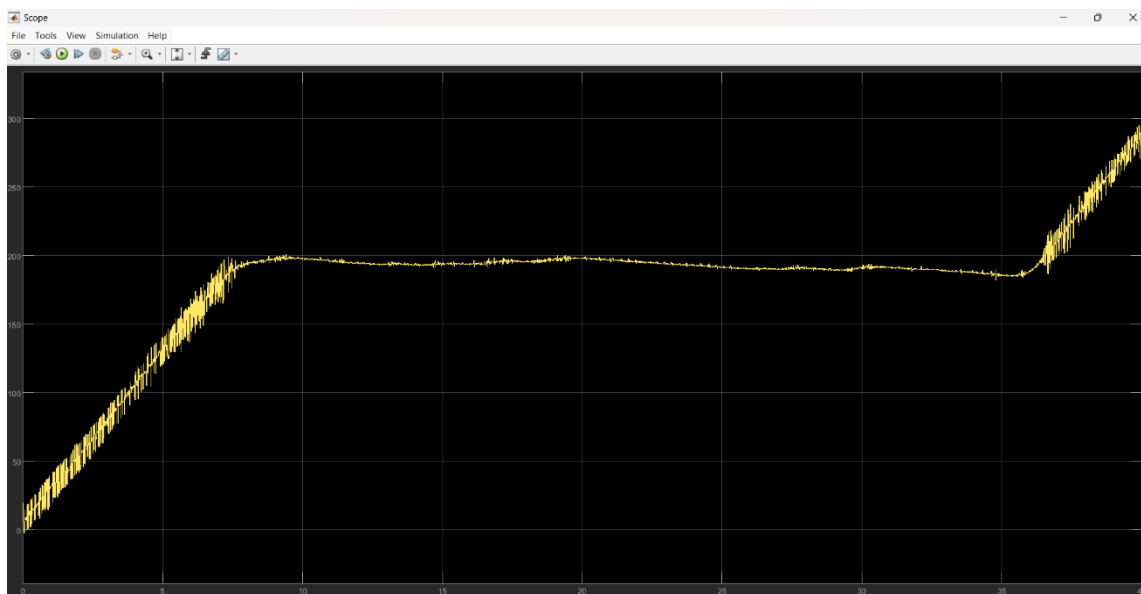
Results after suddenly applying this change (shutting off the power supply):

Error reading (input of the controller):



-We can notice a noise in the first 10 seconds is because of the nature of the fan, which somewhat takes time to push the ball above (error is high and oscillating).

Output of the controller (PWM input):

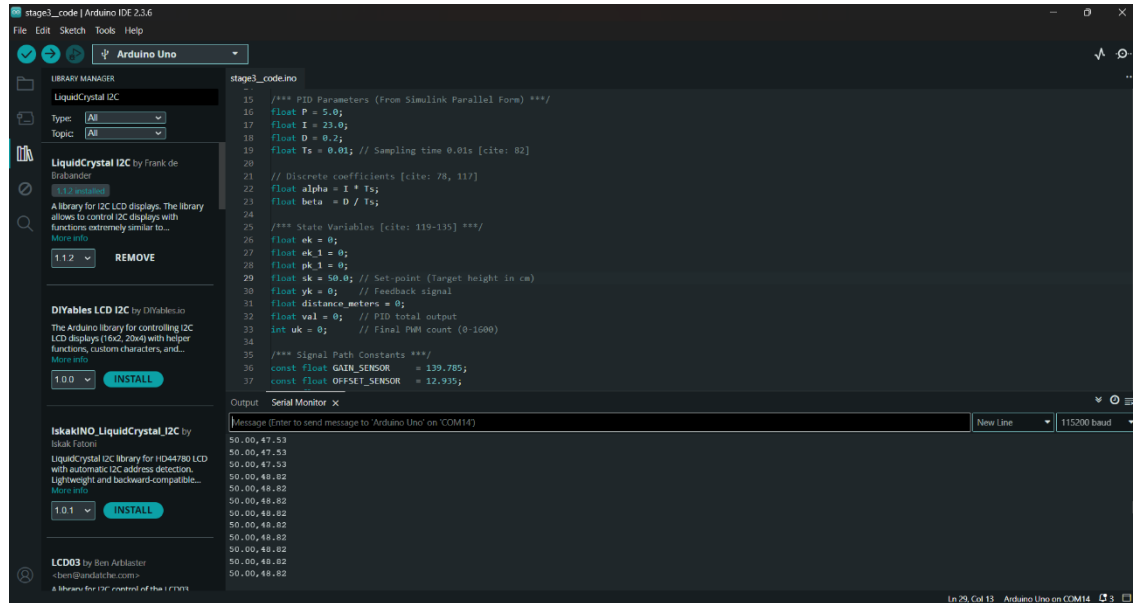


So by this test, we can confirm that the controller works normally and feel the controlling process.

Arduino Code Implementation:

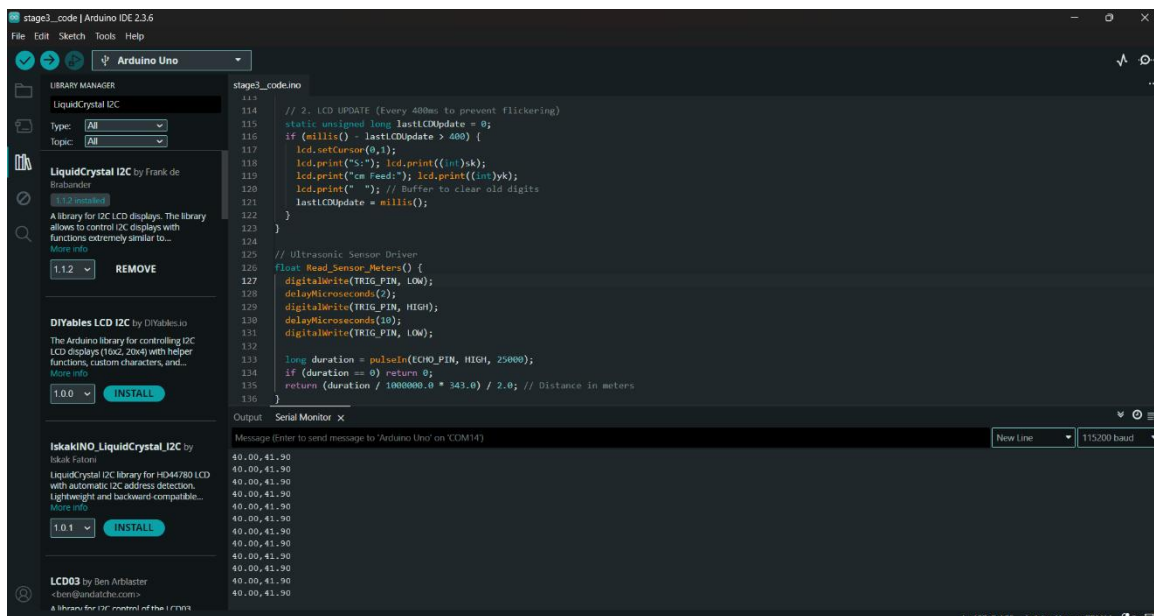
We successfully operated the system by using Arduino code, and it will be uploaded with the delivered files.

Here is the 50 cm set point output:



We can notice from the serial monitor that the output is oscillating around the 50 cm target position, which is the required.

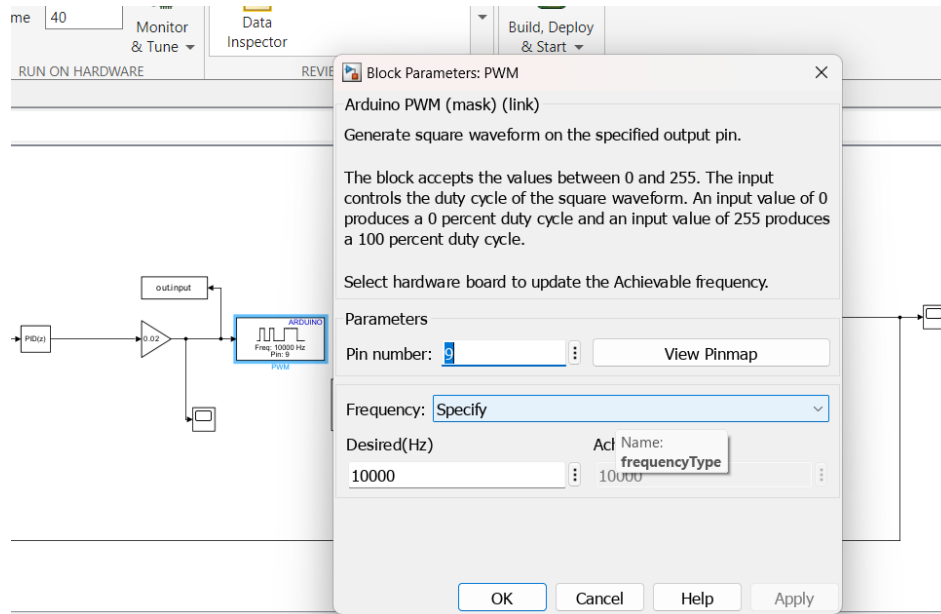
40 cm set point output:



PWM frequency change:

We can see that a small change in PWM will affect the position of the ball, but the ball is somehow slow to respond to the changes.

To solve this problem, we raised the frequency of the PWM to 10000 HZ at pin 9 (PWM).



Finally, we attached an LCD to our hardware system to make the combined system professional and to get clear results. Also, the video of our operating system with Arduino code and the LCD displaying the results will be uploaded with the other files.

